

PENETRATION TESTING REPORT

FOOTPRINTING, OSINT & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

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Program / Batch	Cybersecurity Program - Networkwalks
Date	21 August 2026
Modules Completed	W2-PM1 – Footprinting & Reconnaissance with Kali Linux Tools W2-PM3 – Footprinting & Reconnaissance with Maltego W2-PM4 – Footprinting & Reconnaissance with theHarvester W2-PM5 – Network Scanning with Zenmap
Client / Target	networkwalks.com microsoft.com (OSINT) own local LAN
Permission Secured?	Yes
Phases Covered	Phase 1: Recon & Footprinting Phase 2: Scanning Phase 3-5: In Progress

1. Liability Disclaimer

I have performed these activities only on the systems and domains where I had secured written permission, on public OSINT sources, or on devices/networks that I own myself. All these materials are for education and research purposes only. Do not use anything from here to break the law. The instructor, the authors and Networkwalks are not responsible for what you do with this knowledge. Every action taken is my own responsibility. Misuse can lead to criminal charges, heavy fines, loss of employment and a permanent record. In most countries, unauthorised access is a crime even when nothing is damaged.

2. Introduction

This report covers footprinting the networkwalks.com domain using multiple Kali Linux tools (W2-PM1), OSINT link-analysis with Maltego (W2-PM3), e-mail/subdomain harvesting with theHarvester (W2-PM4), and scanning my own local host-only network with Zenmap (W2-PM5). Together these activities show how an attacker moves from gathering public information to visually mapping relationships between entities, and finally to identifying live hosts on a network. This is the Week 2 part of my ongoing internship program at Networkwalks.

All commands were run in Kali Linux (footprinting, OSINT) and Zenmap was used to scan my own VirtualBox host-only adapter (scanning). Every step below includes the exact command used, the result observed, a screenshot as evidence, and a short note on why the finding matters from an attacker's point of view.

3. Tools Used

Tool	Purpose
Kali Linux & Windows	Operating systems used for reconnaissance, OSINT and scanning activities
WHOIS	Find domain registration details (owner, dates, name servers)
whatweb	Fingerprint web technologies (server, CMS, plugins, IP)
nslookup	Resolve the domain name to its IP address using DNS
curl -I	Read the HTTP response headers of the website
wafw00f	Detect whether a Web Application Firewall protects the site
dnsrecon	Enumerate DNS records (NS, MX, SPF, TXT, SRV) and check name server versions
Maltego (CE)	Visual link-analysis / OSINT transforms to map domain, e-mail and infrastructure links
theHarvester	Passive OSINT harvesting of e-mail addresses and subdomains from public/search sources
Zenmap (Nmap GUI)	Scan the local subnet to find live hosts and generate a topology map

4. Activities Performed

4.1 Footprinting & Reconnaissance — networkwalks.com

I performed reconnaissance against the networkwalks.com domain using six Kali Linux command-line tools: WHOIS, WhatWeb, Nslookup, Curl, Wafw00f and DNSRecon. Each tool was used to collect a different type of information about the target.

First, I used WHOIS to obtain publicly available domain registration information. The results showed the domain is registered with GoDaddy.com, LLC, created on 06 November 2019 and set to expire on 06 November 2027, with name servers ns6135.hostgator.com and ns6136.hostgator.com, and all four client transfer/delete/renew/update prohibited locks enabled.

I then used WhatWeb to identify technologies used by the website. The results identified WordPress 7.1 and WP Download Manager 3.3.58, running on Apache with Bootstrap 7.1 and jQuery 3.7.1, and exposed the contact address info@networkwalks.com and server IP 192.232.216.135.

Using Nslookup, I resolved the domain name to its IP address via Google's public resolver (8.8.8.8). The result identified 192.232.216.135, matching the IP seen in WhatWeb.

I used Curl with the -I option to inspect the HTTP response headers. This confirmed the WordPress REST API endpoint (/wp-json/), a permissions-policy header referencing several third-party token/CAPTCHA providers, a __wpdm_client session cookie, and caching headers (x-nginx-cache, x-endurance-cache-level) that reveal the hosting stack.

Next, I used Wafw00f to determine whether a Web Application Firewall was protecting the website. The result identified ModSecurity (SpiderLabs) after 2 requests.

Finally, I used DNSRecon to enumerate DNS records. The results identified the SOA/NS records, disclosed the BIND version (9.16.23-RH) running on both name servers, an MX record (mail.networkwalks.com), SPF and Google site-verification TXT records, and 8 SRV records for _autodiscover pointing to cpanelmaildiscovery.cpanel.net across multiple IPs.

4.2 OSINT Link Analysis — Maltego

I loaded networkwalks.com as a Domain entity in Maltego Community Edition and ran the 'To Emails [Search Engine]' transform under the Footprint machine set. The graph returned a single linked e-mail entity, info@networkwalks.com, visually confirming the same contact address that WhatWeb exposed and demonstrating how Maltego correlates OSINT data into a link graph for faster analysis.

4.3 OSINT Harvesting — theHarvester

As a separate OSINT training exercise against the public domain microsoft.com, I ran theHarvester with the Baidu search source. The tool returned 4 e-mail addresses (including contact@, support@ and info@microsoft.com) and 16 hosts/subdomains such as account.microsoft.com, graph.microsoft.com and security.microsoft.com, all gathered passively from public search-engine data.

I then re-ran theHarvester against microsoft.com with the -b all flag to query every configured source. The output showed that most premium sources (bevigil, Bitbucket, bufferoverun, BuiltWith, Brave Search, Censys, criminalip, Dehashed, and others) failed because no API key was configured in api-keys.yaml. This highlights that a default theHarvester installation only realises a fraction of its OSINT potential until it is properly licensed and configured.

4.4 Network Scanning with Zenmap

For the final activity, I used Zenmap to perform network discovery on my own local VirtualBox host-only network. I entered the target 192.168.56.1 (the host-only adapter gateway) and ran the built-in Ping Scan profile (nmap -sn 192.168.56.1).

The scan confirmed that the host 192.168.56.1 is up, and Nmap completed the scan of 1 IP address (1 host up) in 0.65 seconds. I then opened the Topology tab in Zenmap and exported the network graphic, which shows a single reachable node (192.168.56.1) connected to my local machine (localhost) — the expected result for a host-only virtual adapter with no other devices attached.

5. Risk Analysis / Impact

Based on the information collected during the footprinting, OSINT and network scanning activities, I identified the following potential risks.

#	Risk / Finding	Evidence / Observation	Potential Impact	Level
1	Web technology information exposed	whatweb identified WordPress 7.1, Bootstrap 7.1 and WP Download Manager 3.3.58	Attackers may target known software versions requiring further security review	Medium
2	Server IP address identifiable	nslookup resolved networkwalks.com to 192.232.216.135	Provides information about the network location of the web service	Low
3	HTTP technical information exposed	curl -I returned response headers and exposed the /wp-json/ REST API endpoint and a session cookie name	May assist technology fingerprinting and further enumeration	Low
4	WAF technology identifiable	wafw00f identified ModSecurity (SpiderLabs) protecting the site	Reveals information about the web application's security architecture	Low
5	DNS infrastructure and BIND version exposed	dnsrecon enumerated NS/MX/TXT/SRV records and disclosed BIND version 9.16.23-RH on both name servers	DNS/software version information can help build a broader infrastructure profile and identify outdated services	Medium
6	Public e-mail address harvested	Maltego 'To Emails [Search Engine]' transform returned info@networkwalks.com from the domain entity	Harvested addresses can be used for phishing or social-engineering pretexting	Medium
7	Subdomains and public e-mails discoverable via OSINT	theHarvester (Baidu source) returned 4 e-mail addresses and 16 hosts/subdomains for the microsoft.com training target	A larger attack surface (subdomains) and contact addresses can be mapped without touching the target directly	Medium

8	OSINT enumeration limited by missing API keys	theHarvester full run (-b all) failed for bevigil, Bitbucket, bufferoverrun, BuiltWith, Brave, Censys, criminalip and Dehashed sources due to missing API keys	Shows current tool configuration under-reports exposure; a fully-keyed run would surface more data	Low
9	Live host visible on local network	Zenmap ping scan (nmap -sn 192.168.56.1) confirmed the host-only adapter gateway 192.168.56.1 is up	Confirms the local test network is reachable; on a real LAN this step would reveal all live devices	Low

Risk level key: **Critical** **Medium** **Low**

The risks above are observations from the footprinting, OSINT and scanning exercises, not confirmed vulnerabilities. These practical exercises primarily involved information gathering and host discovery — no exploitation or vulnerability validation was performed. Therefore, the presence of information such as a software version, IP address, harvested e-mail or DNS record does not by itself mean that the system is vulnerable. Further authorized security testing would be required to confirm any actual vulnerability.

6. Recommendations

Based on the observations from these activities, I recommend the following security improvements:

- Review publicly exposed technology information — regularly audit what CMS, plugin and server version details are visible to WhatWeb-style fingerprinting.
- Keep software updated — CMS platforms, plugins (e.g. WP Download Manager) and other web technologies should be patched and reviewed against current advisories.
- Review HTTP headers — remove or minimise unnecessary technical details (cookie names, cache headers, endpoint links) in HTTP responses.
- Harden DNS server disclosure — suppress BIND version banners on authoritative name servers and review DNS records regularly.
- Properly configure and monitor the WAF — keep ModSecurity enabled and tuned, since it already blocks naive attacks.
- Limit public e-mail exposure — use role-based or form-based contact methods where possible to reduce what OSINT tools like Maltego and theHarvester can harvest.
- Perform regular internal network discovery — periodically scan owned networks to identify active/unexpected devices.
- Investigate unknown devices — any unexpected host discovered during scanning should be investigated and verified.
- Maintain documentation — keep network topology and OSINT-exposure findings documented and updated regularly.
- Perform security testing with authorization — reconnaissance, OSINT and scanning should only be performed against systems/domains where appropriate permission has been granted, or on public information for training purposes.

7. Conclusion

During Week 2 of my Cybersecurity & Ethical Hacking internship, I completed practical activities covering footprinting, OSINT link analysis, e-mail/subdomain harvesting and network scanning.

In the footprinting activity, I used six Kali Linux tools to collect information about the networkwalks.com domain. I learned how WHOIS can provide domain registration information, WhatWeb can identify web technologies, Nslookup can resolve domain names, Curl can inspect HTTP headers, Wafw00f can identify a WAF, and DNSRecon can enumerate DNS records and expose server-software versions.

In the OSINT activities, Maltego showed me how link-analysis graphs can visually correlate a domain to harvested e-mail addresses, while theHarvester demonstrated how quickly subdomains and contact addresses can be gathered from public search engines — and how much more could be found once premium API sources are configured.

In the network scanning activity, I used Zenmap to scan my own host-only virtual network, confirm which host was live, and generate a network topology graphic as evidence.

These exercises showed me that information gathering is a critical first phase of any cybersecurity engagement. Even before attempting to exploit a system, a security professional can learn a significant amount about an environment by carefully analysing publicly available information, DNS/HTTP responses and OSINT sources. I also learned that findings should be documented clearly — what was performed, what was discovered, what it means, what risk it may create, and what can be done to reduce that risk — and that reconnaissance, OSINT and scanning must always be performed within an authorized scope.

8. Evidences Collected

8.1 WHOIS Lookup — networkwalks.com

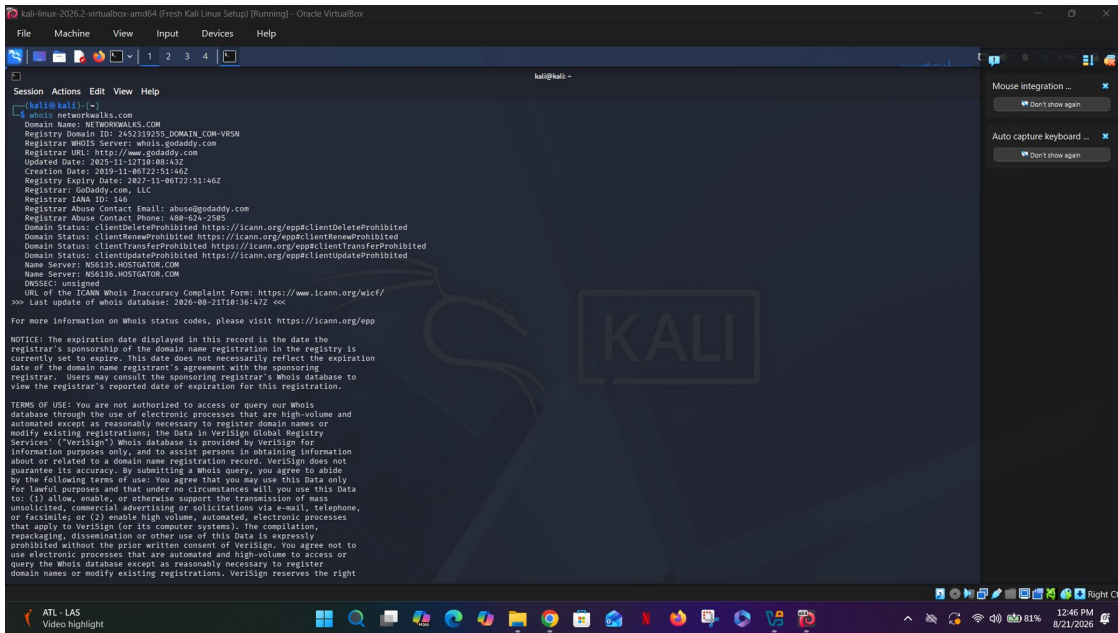


Figure 1: whois networkwalks.com — registrar, dates and name server details

8.2 WhatWeb Fingerprinting — networkwalks.com

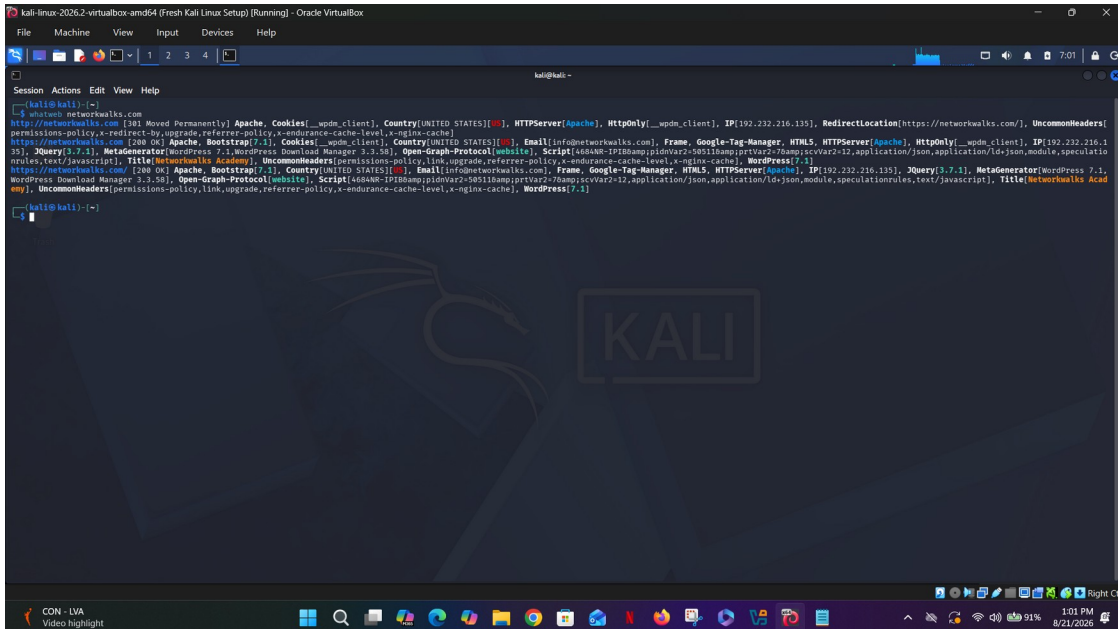


Figure 2: whatweb networkwalks.com — WordPress, Bootstrap, jQuery, IP and contact e-mail

8.3 Nslookup — networkwalks.com

```
(kali㉿kali)-[~]
$ nslookup networkwalks.com
Server:      8.8.8.8
Address:     8.8.8.8#53

Non-authoritative answer:
Name:   networkwalks.com
Address: 192.232.216.135
```

Figure 3: nslookup networkwalks.com — resolves to 192.232.216.135

8.4 HTTP Headers — curl -I

```

kali@kali:~$ curl -I https://networkwalks.com
HTTP/2 200
permissions-policy: private-state-token-redemption(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcapcha.com"), private-state-token-issuance(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcapcha.com")
link: <https://networkwalks.com/wp-json/>; rel="https://api.w.org/", <https://networkwalks.com/wp-json/wp/v2/pages/53/>; rel="alternate"; title="J50M"; type="application/json", <https://networkwalks.com/>; rel="shortlink"
referrer-policy: no-referrer-when-downgrade
x-endurance-cache-level: 0
x-gtm-cache: WordPress
content-type: text/html; charset=UTF-8
date: Fri, 21 Aug 2020 11:08:59 GMT
server: Apache

kali@kali:~$

```

Figure 4: `curl -I https://networkwalks.com` — response headers and `/wp-json/` endpoint

8.5 WAF Detection — wafw00f

```
(kali㉿kali)-[~]  
$ wafw00f networkwalks.com  
  
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    ??  
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          .'.  (. ) ;' ,.'     "      "'  
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       -". , ..'_.,,)_(..,( .' )_-'_'_')(. _..(''  
       |_____|_____||_____||_____||_____||  
  
           ~ WAFW00F : v2.4.2 ~  
           ~ Sniffing Web Application Firewalls since 2009 ~  
  
[*] Checking https://networkwalks.com  
[+] The site https://networkwalks.com is behind ModSecurity (SpiderLabs) WAF.  
[~] Number of requests: 2  
  
(kali㉿kali)-[~]  
$ █
```

Figure 5: wafw00f networkwalks.com — ModSecurity (SpiderLabs) detected

8.6 DNS Enumeration — dnsrecon

```
[kali@kali]~$ dnsrecon -d networkwalks.com
2026-08-21T07:13:27.800697-0400 INFO Starting enumeration for domain: networkwalks.com
2026-08-21T07:13:27.801181-0400 INFO std: Performing General Enumeration against: networkwalks.com...
2026-08-21T07:13:28.182559-0400 ERROR No answer for DNSSEC query for networkwalks.com
2026-08-21T07:13:28.381203-0400 INFO SOA ns6135.hostgator.com 50.87.144.87
2026-08-21T07:13:28.640650-0400 INFO NS ns6135.hostgator.com 192.232.216.131
2026-08-21T07:13:28.966206-0400 INFO Bind Version for 192.232.216.131 "9.16.23-RH"
2026-08-21T07:13:28.966406-0400 INFO NS ns6135.hostgator.com 50.87.144.87
2026-08-21T07:13:29.286115-0400 INFO Bind Version for 50.87.144.87 "9.16.23-RH"
2026-08-21T07:13:29.813378-0400 INFO MX mail.networkwalks.com 192.232.216.135
2026-08-21T07:13:29.888676-0400 INFO A networkwalks.com 192.232.216.135
2026-08-21T07:13:30.420045-0400 INFO TXT networkwalks.com v=spf1 +a -mx +ip4:50.87.144.87 +include:websitelwelcome.com -all
2026-08-21T07:13:30.490440-0400 INFO TXT networkwalks.com google-site-verification=rr04eRmqH0wY3Xemm1zDNVK4q75X-Ij-mjgEeg-USYi
2026-08-21T07:13:30.824067-0400 INFO Enumerating SRV Records
2026-08-21T07:13:32.981727-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanel.emailldiscovery.cpanel.net 184.94.204.14 443
2026-08-21T07:13:32.981908-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanel.emailldiscovery.cpanel.net 184.94.203.14 443
2026-08-21T07:13:32.981960-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanel.emailldiscovery.cpanel.net 184.94.204.15 443
2026-08-21T07:13:32.981998-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanel.emailldiscovery.cpanel.net 184.94.203.11 443
2026-08-21T07:13:32.982034-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanel.emailldiscovery.cpanel.net 184.94.204.9 443
2026-08-21T07:13:32.982072-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanel.emailldiscovery.cpanel.net 184.94.204.11 443
2026-08-21T07:13:32.982108-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanel.emailldiscovery.cpanel.net 184.94.203.15 443
2026-08-21T07:13:32.982170-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanel.emailldiscovery.cpanel.net 184.94.203.9 443
2026-08-21T07:13:32.983399-0400 INFO R Records Found
2026-08-21T07:13:32.983854-0400 INFO Completed enumeration for domain: networkwalks.com
```

Figure 6: dnsrecon -d networkwalks.com — NS, MX, TXT and SRV records

8.7 OSINT Link Analysis — Maltego

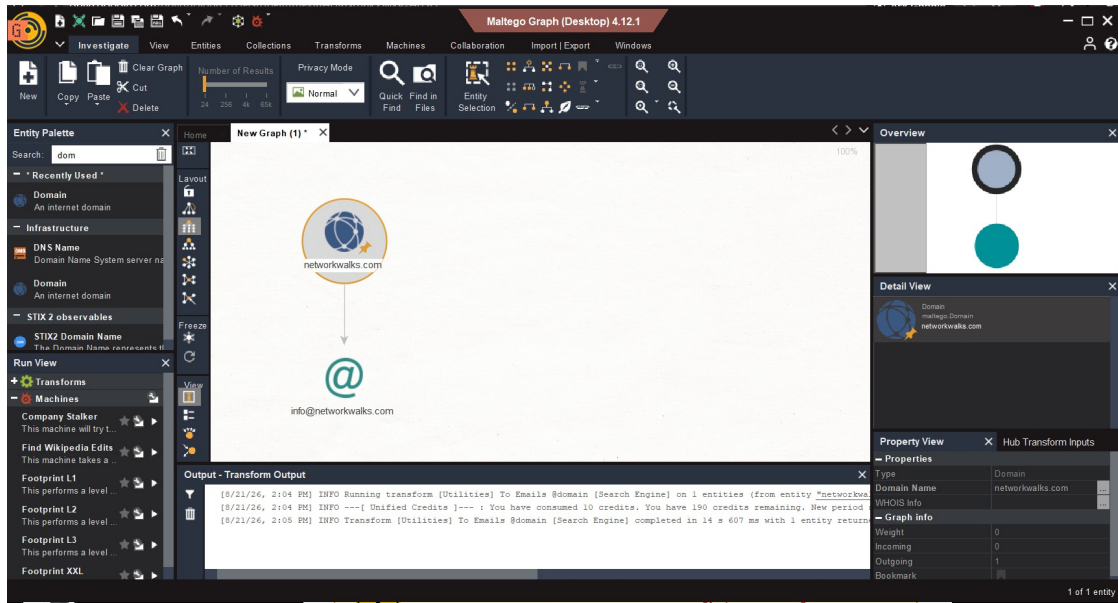


Figure 7: Maltego graph — networkwalks.com Domain entity linked to info@networkwalks.com

8.8 OSINT Harvesting — theHarvester (Baidu source)

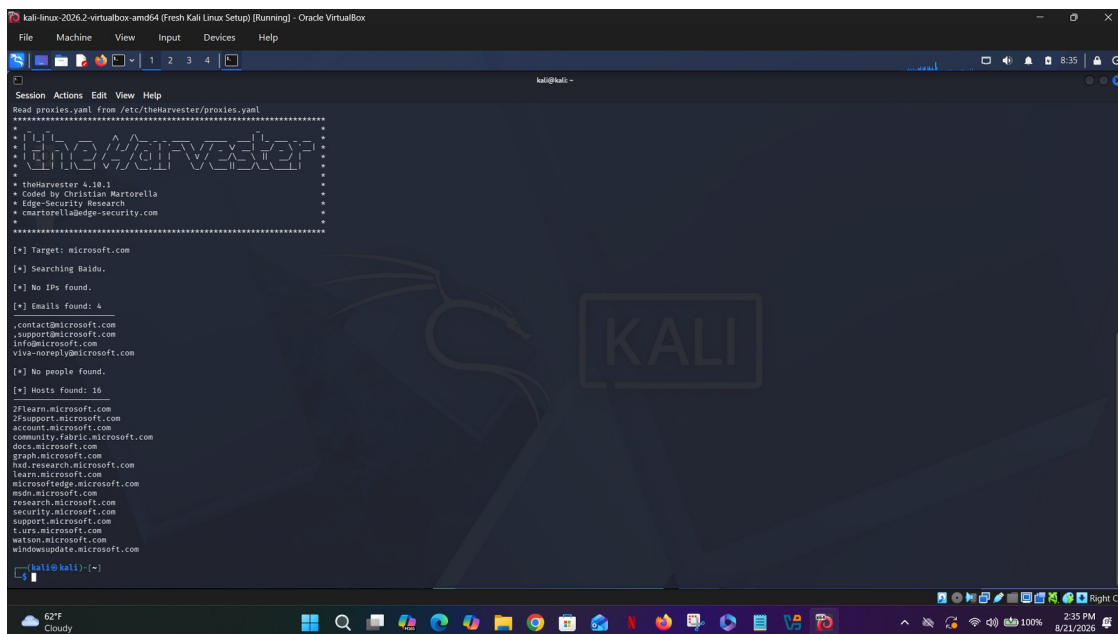


Figure 8: theHarvester -d microsoft.com — 4 e-mails and 16 hosts found via Baidu

8.9 OSINT Harvesting — theHarvester (all sources)

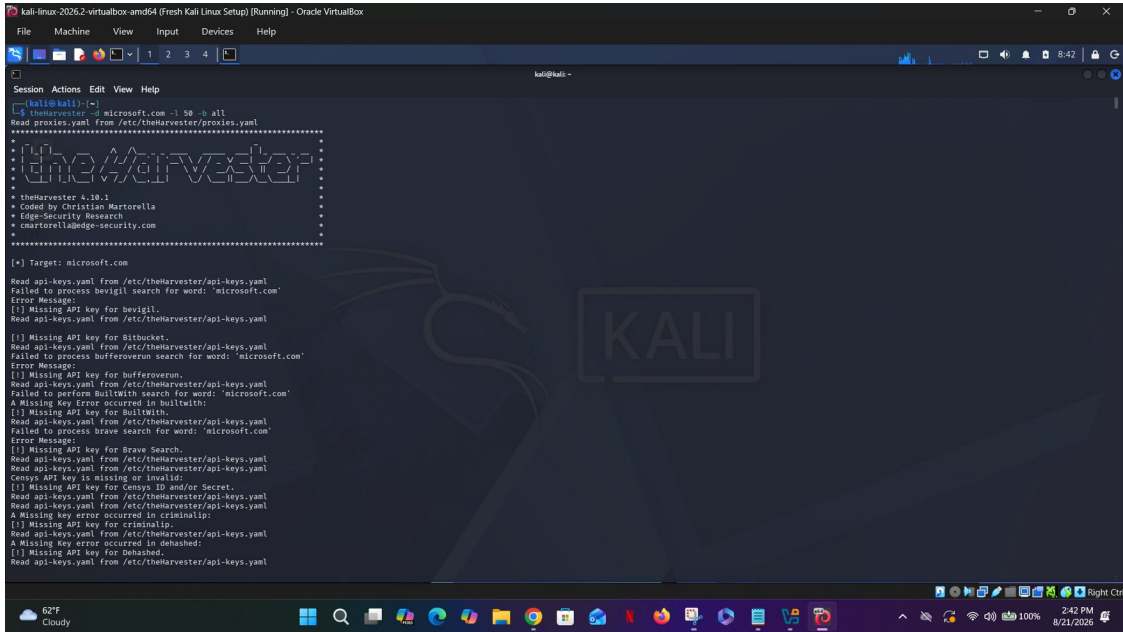


Figure 9: theHarvester -d microsoft.com -l 50 -b all — most sources fail without configured API keys

8.10 Network Scan — Zenmap Ping Scan

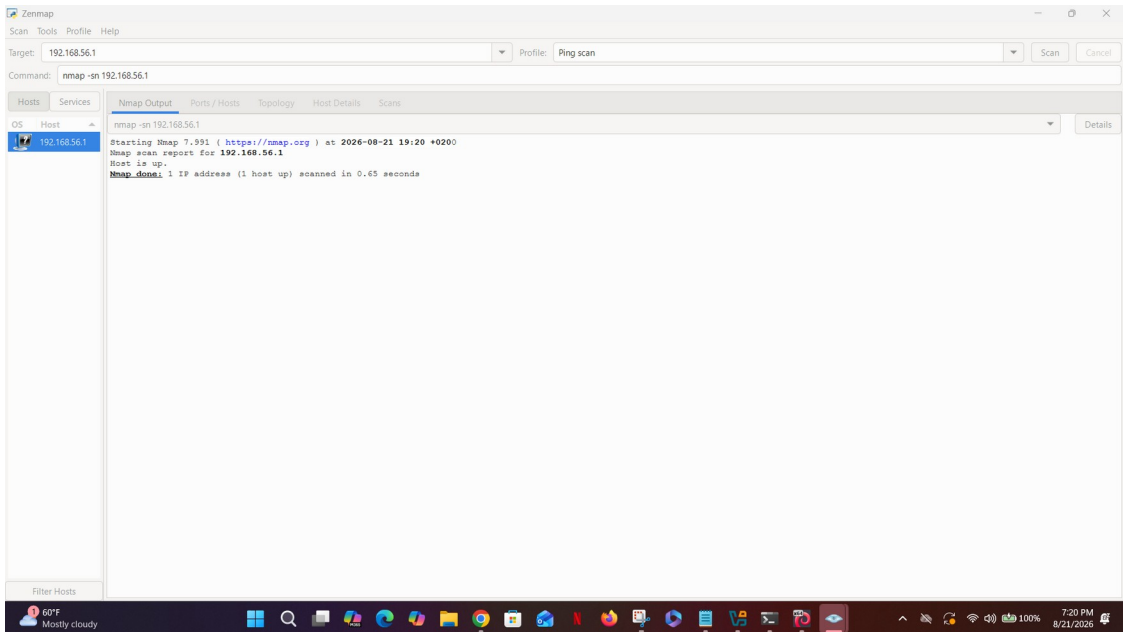


Figure 10: Zenmap — nmap -sn 192.168.56.1, 1 host up in 0.65 seconds

8.11 Network Topology — Zenmap

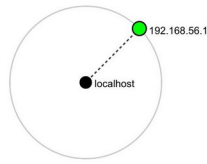


Figure 11: Zenmap topology graphic — 192.168.56.1 linked to localhost

- End -

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Project Information

Program Name: Cybersecurity Program at Networkwalks | Week: 02 | Repository: GitHub